

a piece of code or text. The left pane of your notebook has buttons that let you delete, insert or move cells around. The default entry mode into a cell is 'code'; so you can go right ahead and enter the 'Hello world' code snippet. As you hit `<Enter>`, the cell grows in size, letting you enter more code. You have two modes to execute your code. A `<Ctrl> + <Enter>` executes your code in-place. Which means you could keep editing the same piece of code to say 'Hello' to all your friends by hitting `<Ctrl> + <Enter>` to see the output in the same cell. There is no execution history, and the cell holds the last piece of code you entered along with its output.

A `<Shift> + <Enter>` lets you run the code in the cell and opens a new cell for input. This way you have a history of cell executions that might be leading up to some programming goal you have set for yourself. Don't worry, you can always delete the cells that do not fit in.

What is code without documentation? You can switch a cell from the code to the text mode by selecting 'markdown' from the left pane. This option lets you enter text in a cell using the markdown format, which is similar to the wiki syntax. The markdown is rendered as HTML.

IPython lets you embed graphs and plots in line with your notebook. You could use *Matplotlib* or GNU Octave to draw graphs that integrate seamlessly with your IPython notebook. There is tab completion available, too.

Then there are the supposed magic commands. These get you some 'meta' functionality. For instance:

```
%timeit -n10 -r5 print "Hello World"
```

...would time the statement for you over 10 iterations, taking the best of five. A `%lsmagic` will get you all the magic commands available to you. To get help on a magic command, you just need to issue the magic command appended by a `?` mark. A `%bookmark?` command would give you help on IPython's bookmark command, for instance.

For the scientific types, there is support for embedding graphs as mentioned earlier. One way to learn how to do this is to issue the `%pylab inline` command before trying out the terrific tutorials on the IPython site.

By now you might also have noticed the *Print* command in the left pane, and have also tried printing your notebook. This works very much like printing any HTML page, which means you can also save your notebook to a PDF file.

## Usage scenarios

For a student-learner, leveraging the features of IPython is quite straightforward. You can carry on learning new recipes and keep a live notebook that you keep updating. Or you can

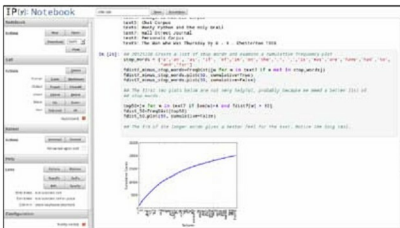


Figure 1: An open notebook with code and output

print your notebook of recipes and use it as a reference, both offline and online.

For a teacher, IPython can be a great way not only to teach but also to hand out and evaluate assignments. The interactivity of IPython lets it fit very well into today's e-classrooms equipped for digital media. Since IPython notebooks can be readily shared, a teacher can post an assignment as an IPython notebook. Students can import this notebook to their own IPython session, work out their assignments and upload them to a designated location. Since the notebook is 'live', the teacher or instructor can actually interact with the student's code for a fair evaluation and also insert comments (remember you can insert cells and text into a notebook).

As a scientist or student trying to make sense of data, you might have used Numpy or Matplotlib. IPython gives you a convenient environment to work with data by letting you experiment to your heart's content and also embed plots, graphs or other rich data in-line in your notebook, and share them with co-workers. It also lets you use LaTeX to typeset any math expression worth its value, using Mathjax.

## Create and share

IPython is also a great way to create and share live tutorials and instructional material. So now that you know about IPython, I trust you will never go back to learning or teaching programming the 'old fashioned' way. 

## References

- [1] [www.python.org](http://www.python.org) - The IPython website
- [2] [www.daringfireball.net/projects/markdown-explains](http://www.daringfireball.net/projects/markdown-explains) - markdown syntax

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The author enjoys all things FOSS, especially learning and writing about productivity tools.